

Handoff #4: airline arm 0 complete, first in-loop run

Everything since handoff #3 (commit 4b82f58): every affordable decider replayed on all 1,076 airline states, a round of accounting fixes that changed some headline numbers, and the first real in-loop run (arms B, C, E on airline, all local, \$0).

TL;DR

- Airline arm 0 is done for 7 deciders (Jev, GPT-OSS-20B, Qwen3-8B local, Kev-0.8B/4B/9B, cascade), 1,076 states × 5 repeats. Jev leads clearly: strict 0.814, vs GPT-OSS-20B 0.672 and every local model at or below 0.40. Jev is also the fastest (p50 0.18 s) and cheapest API decider.
- Every paired gap is now significant on ~1,040 states: Jev – GPT-OSS +15.5 pts, Jev – Kev-9B +40.7 pts, Kev-9B – Kev-4B +8.0 pts. (Handoff #3's -12.2 pts for Kev-4B vs 9B was on 662 states; the direction held, the size shrank.)
- First in-loop run (U4), airline, G = user = Qwen3-8B, 750 episodes, 0 errors. Reward: B 0.308 (G decides, then fills) > C 0.196 (G alone) > E 0.048 (Kev-4B decides). Kev fails in the loop the same way it fails in arm 0.
- Accounting fixes changed numbers: a spending guard after the account ran dry mid-run, errors split into decider failures vs infra, retries of decider failures stopped, energy pooled across resumes, and cascade \$ and J corrected (mock cascade \$ had been overstated 2.5×).
- Still blocked on credits: the frontier ceiling decider (U3), the remaining 4,005 cascade and 719 GPT-OSS calls, and arms A and D in the loop. Router and cache-guard decision points haven't been started.

Your first 5 minutes

```
git pull    # airline replay files for every decider, in-loop harness, this PDF
less results/replay/airline/report/summary.md # arm 0, all 7 deciders
less DECISIONS.md # 13 new rows dated 2026-09-25/26: every fix and why
less PLAN.md  # G1-G4, U1, U2 ticked; U3/U4 partial
```

In-loop results (results/inloop/) are git-ignored (.gitignore shares only states and replay). They're on Gilbreth scratch in the repo checkout. The numbers are copied into DECISIONS.md and into this PDF.

1 · Where we are

Decision points x deciders

	<i>Ceiling (frontier LLM)</i>	<i>Baseline (Qwen3-8B)</i>	<i>Floor (classifier)</i>	<i>Jev (under test)</i>
Router	TODO	TODO	TODO	TODO
Cache guard	TODO	TODO	TODO	TODO
Agent control	PART — gpt-5.6 generated the airline states; not yet replayed as a decider (U3)	DONE — arm 0 on mock + airline; G in the in-loop run	TODO — Kev-0.8B is the cheapest local point, but it's not a classifier	DONE — arm 0 on mock + airline

Agent control is the only decision point with data: 2 of 12 cells done, 1 partial.

Experimental arms (agent control, tau2 airline)

<i>Arm</i>	<i>What</i>	<i>Status</i>
0	each decider -> none (decision quality + cost)	PART — 7 deciders done; frontier LLM missing; cascade 4,005 calls and GPT-OSS 719 still to run
A	Jev (API) -> G	TODO — one-task vertical slice only; the full run needs credits
B	G choice call -> G	DONE — reward 0.308, local G = Qwen3-8B
C	G, one call (production default)	DONE — reward 0.196, local G
D	small LLM -> G	TODO — needs credits
E	Kev (local GPU) -> G	DONE — reward 0.048, local G

B/C/E were run with a free local G (Qwen3-8B). The planned runs with a frontier model as G, which the paper's headline needs, haven't happened.

PLAN.md checklist

- Done: G1 (Kev-4B/9B mock), G2 (compute-node egress, from existing evidence), G3 (Qwen3-8B local), G4 (airline repeats + Kev-0.8B), U0, U1 (lenient labels), U2 (airline states).
- Partial: U3 (Jev + GPT-OSS-20B on airline; the frontier LLM is missing), U4 (harness built, B/C/E run with local G; A/D and frontier-G runs not done).
- Not started: U5 (paper tables and figures from all arms).

2 · Airline arm 0: every affordable decider

1,076 states from reward=1 episodes (the 689 in handoff #3, plus the fill run for tasks 38–49), 5 repeats each. Each (state, repeat) counts once, as its first attempt that wasn't an infra error.

Decider	Strict [95% CI]	Lenient	p50 latency	\$ / decision	J / dec. (gross)	Failures
Jev (API)	0.814 [0.791, 0.836]	0.826	0.176 s	1.73e-4	hosted	0.0%
Cascade Kev-4B → GPT-OSS, t=0.9 *	0.669 [0.634, 0.704]	0.685	2.100 s	3.28e-4	132 (Kev stage)	7.2%
GPT-OSS-20B (API) *	0.672 [0.647, 0.696]	0.688	1.161 s	3.32e-4	hosted	9.2%
Kev-9B (A100)	0.401 [0.371, 0.431]	0.546	0.548 s	local	170.5	3.2%
Qwen3-8B (A100, vLLM)	0.396 [0.368, 0.425]	0.442	0.656 s	local	211.2	0.0%
Kev-4B (A100)	0.321 [0.294, 0.350]	0.445	0.439 s	local	132.4	3.2%
Kev-0.8B (A100)	0.273 [0.247, 0.300]	0.418	0.218 s	local	47.2	3.3%

* Incomplete. The OpenRouter balance ran out mid-run: the cascade has 4,005 calls still missing (729 of 1,076 states seen), GPT-OSS-20B has 719. Kev failures are its hard 8,192-token input limit on the longest states ("branch too long"), which counts as a coverage gap. GPT-OSS failures are replies that came back without the forced tool call.

Paired significance (cluster bootstrap over states)

Comparison (airline)	States	Diff	95% CI	Significant?
Jev vs GPT-OSS-20B	1,073	+15.5 pts	[+13.2, +17.8]	Yes
Jev vs Kev-9B	1,042	+40.7 pts	[+37.4, +44.1]	Yes
Kev-9B vs Kev-4B	1,042	+8.0 pts	[+5.0, +10.8]	Yes
Cascade vs Kev-4B	729	+32.7 pts	[+28.4, +37.2]	Yes (incomplete run)

Reading it: on airline, Jev is the best decider by a wide margin: more accurate than GPT-OSS-20B, 6.6× faster, 1.9× cheaper, and the best calibrated (ECE 0.029). The local models tell a separate energy story. Kev-0.8B gives the most accuracy per joule (0.58 %/J vs Kev-4B 0.24), and Qwen3-8B costs the most energy for accuracy similar to Kev-9B. Qwen3-8B (thinking off) is badly overconfident: ECE 0.60 with mean confidence 0.996. Its errors are real judgment errors, e.g. 508 cancel_reservation choices, often before any lookup.

3 · Accounting fixes (some moved numbers)

Found in a full review of the uncommitted work before the next paid run. Each has a dated row in DECISIONS.md with the reasoning.

The account ran dry mid-run, again

The airline replay was launched with ~\$2.80 left for ~\$4.50 of calls, and nothing checked the balance. It ended at \$10.15 used of \$10: 719 GPT-OSS and 1,794 cascade calls failed with 402 `in_flight_budget_exhausted`. Fix: `replay_decisions.py` now estimates the run's \$ from earlier calls, refuses to start above `--max-usd` or the live balance ($\times 1.25 + \$0.25$ headroom, because OpenRouter pre-authorizes in-flight requests), and stops on the first 402. `--dry-run` prints the estimate only.

Errors: decider failures vs infra

- New `src/dwg/errors.py`. Infra errors (402, 429, provider 5xx, network) are retried on resume and never count against a decider. Decider failures (no tool call, malformed JSON, input too long) are final.
- Before this, resume retried every errored row, so a stochastic decider could re-roll its own failures until one succeeded, which inflated accuracy. The old "402" in error check would also have stopped all paid calls on a Kev "branch too long: 4020 tokens" message.
- Each (decider, state, repeat) counts once, as its first non-infra attempt. Mock numbers moved by ≤ 3 calls.

Cost and energy

- Cascade \$: a non-escalated call now costs \$0 instead of "unknown". Mock cascade had been reported at $8.8e-5$ \$/decision; the true figure is $3.5e-5$ ($2.5\times$ lower). Airline is unaffected: Kev-4B never reaches 0.9 confidence there, so every call escalates.
- Cascade J now comes from the same run's Kev-4B (132 J on airline), not the hard-coded mock value of 18.33 J.
- Energy blocks are pooled across resumes. Airline Kev-4B 133.4 $\rightarrow$ 132.4 J, Kev-9B 177.3 $\rightarrow$ 170.5 J.
- Price routing: airline GPT-OSS calls were billed ~ 2.7 - $3.4\times$ list price (likely pricier upstreams). Each call now records its provider. Use `--provider-sort` price from the start of new runs; don't switch partway through.
- Lenient accuracy (U1) also accepts a READ lookup before a WRITE reference. Mock: Jev 0.772 $\rightarrow$ 0.877, GPT-OSS 0.852 $\rightarrow$ 0.932.

4 · In the loop: arms B, C, E on airline

New harness: DeciderTau2Agent (the decider picks the tool, G fills in the arguments; arms A/B/D/E share one loop) and tau2's LLMAgent for arm C. Trial t uses seed $2000+t$ in every arm. If the decider fails, G decides that turn, and the turn is logged as a fallback. Job 11823642: 50 tasks $\times$ 5 trials $\times$ 3 arms = 750 episodes, 0 errors. G and the user simulator are both Qwen3-8B (thinking off), \$0.

Arm	Reward [95% CI]	pass ¹	pass ⁵	turns / ep.	decision share of latency	fallback
B: G decides + G fills	0.308 [0.196, 0.428]	0.308	0.200	7.0	32.7%	0.0%
C: G alone	0.196 [0.116, 0.284]	0.196	0.060	11.5	0%	—
E: Kev-4B decides + G	0.048 [0.012, 0.092]	0.048	0.000	10.2	70.4%	2.5%

Why E collapses: Kev's behavior, not a harness bug

106 of 250 E episodes end too_many_errors (B: 0, C: 15). In 92 of them, Kev picks get_user_details before the user has given an ID, and keeps picking it. G, forced to call the chosen tool, fills in sara_doe_496, the example ID in tau2's own tool docstring. The log has 563 "User sara_doe_496 not found" errors. This matches arm 0, where Kev-4B is weak on airline.

B beats C: with an 8B G, separating the decision from the execution helps rather than hurts: +11 pts reward and fewer turns. Whether that holds with a frontier G is the open question the paid runs answer.

Don't quote in-loop latency as model latency. Kev's in-loop p50 is 4.8 s/decision vs ~0.2–0.6 s in arm 0, because 8 concurrent episodes queue on a one-request-at-a-time Kev server that shares the GPU with vLLM. Energy isn't measured in the loop at all. Take latency and J per decision from arm 0; in the loop, report reward, turns and cost shares.

5 · What's next

Blocked on you

Add OpenRouter credits, then pick the frontier model for the ceiling decider (U3). Before launching anything, run `replay_decisions.py --dry-run`: the spending guard will refuse any run the balance can't cover.

Ready to run once credits land

- Finish the airline cascade (4,005 calls) and GPT-OSS-20B (719). Keep default routing for these two: they started on it.
- U3: the frontier LLM on the 1,076 airline states, 5 repeats, with `--provider-sort price`.
- U4 paid: arms A-E with a frontier G, 50 tasks × ≥5 trials. That's the paper's headline comparison.
- Optional, free: Qwen3-8B with thinking on (vLLM `--reasoning-parser qwen3`) as a second baseline setting.

Not started

- Router and cache-guard decision points (8 of the 12 grid cells).
- A real floor decider (a classifier) for agent control.
- U5: paper tables and figures across all arms.

New files this handoff

```
src/dwg/decider_tau2_agent.py # DeciderTau2Agent, TimedLLMAgent (arms A/B/D/E)
scripts/run_inloop.py, scripts/analyze_inloop.py, scripts/inloop_local.slurm
src/dwg/errors.py # is_infra_error, is_out_of_credits
src/dwg/labels.py # strict + lenient accuracy
src/dwg/billing.py, src/dwg/tau2_llm.py
scripts/llm_local_replay.slurm # Qwen3-8B on vLLM, with energy
results/states/airline-fill.jsonl # tasks 38-49 top-up
results/replay/airline/calls-*.jsonl # jev, gpt-oss-20b, qwen3-8b, kev-0.8b, cascade
```

Gotchas we hit this round

- OpenRouter pre-authorizes in-flight requests, so a run can hit 402 before the balance reads \$0. Budget with headroom.
- Never retry a decider's own failures on resume: it silently inflates accuracy. Retry infra errors only.
- Match error codes only where they're reported as a status (`"code": 402`), never as any number in a message.
- Redirected stdout is block-buffered, so a multi-hour job's log stayed empty until exit. `env . sh` now sets `PYTHONUNBUFFERED=1`.
- Kev's first call after load took 8.3 s vs ~0.2 s steady. Free deciders get warmup calls before timing starts.